

2.7 Improper Integrals

Motivation:

Suppose we want to find the area of the region, R , given by $R = \left\{ (x, y) \mid x \geq 1 \text{ and } 0 \leq y \leq \frac{1}{x^2} \right\}$.

The key question here is: does the area under the above curve approach a finite number?

Definition (Improper Integral of Type 1):

1. If $\int_a^t f(x)dx$ exists for every number $t \geq a$, then

2. If $\int_t^b f(x)dx$ exists for every number $t \leq b$, then

Definition: The improper integrals $\int_a^\infty f(x)dx$ and $\int_{-\infty}^b f(x)dx$ are called

Definition: If both $\int_a^\infty f(x)dx$ and $\int_{-\infty}^a f(x)dx$ are convergent, then we define

Example 1 (you try): Determine whether $\int_4^\infty \frac{1}{\sqrt{x}} dx$ is convergent or divergent. If it is convergent evaluate the integral.

Example 2: Evaluate $\int_{-1}^\infty xe^{-x} dx$

Example 3: Evaluate $\int_{-\infty}^{\infty} \frac{1}{1+x^2} dx$.

Definition (Improper integral of Type 2):

1. If f is continuous on $[a, b)$ and discontinuous at b , then

2. If f is continuous on $(a, b]$ and discontinuous at a , then

Definition: If f has a discontinuity at c , where $a < c < b$ (and f is continuous on $[a, c)$ and $(c, b]$) and

both $\int_a^c f(x)dx$ and $\int_c^b f(x)dx$ are convergent, then

Example 4: Evaluate $\int_0^{\frac{\pi}{2}} \tan(x) dx$ if possible.

Example 5 (you try): Consider the following problem: Evaluate $\int_0^3 \frac{1}{x-1} dx$ if possible. Explain what is

wrong with the following solution: $\int_0^3 \frac{1}{x-1} dx = \ln|x-1| \Big|_0^3 = \ln(2) - \ln(1) = \ln(2)$.

Question:

Does $\lim_{t \rightarrow \infty} \int_0^t e^{-x^2} dx$ converge to a finite number?

Theorem (Comparison Theorem):

Suppose that f and g are continuous functions with $f(x) \geq g(x) \geq 0$ for $x \geq a$.

1.

2.

Note: This theorem is commonly called the comparison theorem. Since our book omits this theorem I will omit the proof.

Example 6: Prove that $\int_0^{\infty} e^{-x^2} dx$ is convergent.

Example 7: Gabriel's horn is the solid obtained by rotating the region $R = \left\{ (x, y) \left| x \geq 1 \text{ and } 0 \leq y \leq \frac{1}{x} \right. \right\}$ about the x -axis, find whether or not the surface area is finite.